

Video Transcripts

Week 1.1: Establishing the Core Distinctions between Classical ML, Gen AI and Agentic AI

Video 1: Overview of Classical ML, Generative AI, and Agentic AI

Welcome, everyone. This video sets out to clarify a common source of confusion in organisations: artificial intelligence is not one single capability, but a collection of distinct paradigms that solve different types of problems. While AI is now widely discussed at board level and adopted across many functions, inconsistent business value often stems from treating all AI as the same. The central premise of this session is that understanding the differences between classical machine learning, generative AI, and agentic AI is essential for making effective and responsible strategic decisions.

The first paradigm explored is classical machine learning, often called predictive AI. These systems analyse historical data to identify patterns and estimate future outcomes. Typical use cases include demand forecasting in retail and credit risk assessment in banking. In these scenarios, the system does not make decisions or take action. Instead, it produces probability scores or forecasts that inform human judgement or automated policy rules. The value of predictive AI lies in reducing uncertainty and improving decision quality, particularly at scale. Its limitation is that it relies on historical patterns and can perform poorly when conditions change significantly.

The video then introduces generative AI, which represents a fundamental shift from prediction to creation. Rather than estimating what will happen, generative systems produce new content such as text, code, images, or structured drafts. In practical terms, this enables marketing teams to generate campaign copy, customer service teams to draft responses, and developers to accelerate routine coding tasks. While this leads to significant productivity gains, generative AI does not replace human responsibility. Its outputs may contain errors or bias, and the system has no intent or accountability. As a result, generative AI remains firmly human-in-the-loop, supporting professionals rather than acting independently.

Agentic AI is presented as the third and most advanced paradigm. These systems go beyond prediction and content generation to plan and act in pursuit of defined goals. An agentic system can coordinate multiple steps across tools, such as placing orders, updating systems, and notifying teams, all within predefined boundaries. The progression becomes particularly clear when the paradigms are combined: predictive models identify likely outcomes, generative systems create relevant content, and agentic systems execute actions. As autonomy increases, governance becomes critical, requiring clear rules, oversight, and escalation mechanisms.

The key takeaway is that these paradigms are complementary rather than competing. Classical machine learning improves decisions through prediction, generative AI accelerates knowledge work through creation, and agentic AI enhances execution through coordinated action. Most organisations will use all three, depending on the nature of the task. The leadership challenge is not whether to use AI, but where each paradigm creates the greatest leverage. When organisations clearly distinguish

between prediction, creation, and autonomous action, AI moves from vague experimentation to a structured capability embedded thoughtfully into real workflows.

Video 2: Classical Machine Learning — Prediction as the Core Capability

This video shifts attention away from highly visible generative tools and towards a quieter but economically significant form of AI: classical machine learning. Long before chatbots entered boardroom conversations, predictive systems were already embedded in core business operations, influencing high-stakes decisions across retail, finance and payments. Their impact is substantial precisely because they operate at scale and directly affect financial and operational outcomes.

At its core, classical machine learning works by learning from historical data to estimate what is likely to happen next. It does not generate ideas, produce language, or act autonomously. Instead, it outputs probabilities, forecasts, or risk scores that support human judgement. These predictions help decision-makers manage uncertainty more effectively, but responsibility for action remains firmly with people and established business rules.

Retail demand forecasting provides a clear illustration. Walmart has publicly described using machine learning models to forecast demand across thousands of stores and millions of products. By analysing historical sales, promotions, local events and weather patterns, these systems continuously update forecasts at scale. Even marginal improvements in accuracy can have outsized effects, reducing stock-outs, lowering excess inventory and improving working capital efficiency. Importantly, the model does not decide production or allocation levels. It predicts expected demand, while supply chain teams make the final operational decisions.

Credit risk assessment in banking follows the same predictive logic. JPMorgan Chase uses machine learning models trained on extensive historical data — including repayment behaviour, account activity and economic indicators — to estimate the probability of default. In loan portfolios measured in billions, even small gains in predictive accuracy can materially reduce credit losses and support more precise risk-based pricing. The output remains a risk score, with approval decisions governed by human-defined policies and regulatory frameworks.

Fraud detection across payment networks further reinforces this pattern. Transactions are evaluated in milliseconds and assigned a likelihood of fraud based on past patterns. Low-risk transactions proceed automatically, high-risk ones are blocked, and ambiguous cases are escalated for additional checks. The system handles scale and speed, while human teams focus on judgement-heavy edge cases. Across all these examples, the structure remains consistent: historical data informs prediction, and people determine action.

The effectiveness of classical machine learning depends on the continued relevance of past data. When consumer behaviour shifts or economic conditions change, models must be retrained and recalibrated. Prediction is powerful, but it is inherently grounded in history. For leaders, the lesson is clear: classical machine learning delivers value where reliable data exists and where better prediction directly improves outcomes. It is not creative or autonomous, but a disciplined predictive engine embedded within decision processes. Understanding this boundary helps organisations deploy it responsibly and set realistic expectations within a broader AI strategy.

Video 3: Strengths and Limitations of Classical ML

Classical machine learning is best understood not only by what it does, but by where it delivers exceptional value and where its limitations become visible. These boundaries become clearer when examined through real-world organisational cases rather than abstract definitions.

A strong illustration of classical machine learning operating at scale comes from large logistics networks. UPS uses classical machine learning and optimisation techniques within its ORION routing system to improve delivery efficiency across a global operation handling more than 20 million packages per day. By analysing historical delivery routes, traffic patterns and operational constraints, the system predicts the most fuel-efficient sequence of stops for drivers. The objective is precise and measurable: minimise distance, time and fuel consumption. Even marginal improvements — such as reducing one mile per driver per day — translate into millions of miles saved annually and hundreds of millions of dollars in operating cost reductions. The system does not replace drivers or managers; it enhances route planning with consistent, data-driven precision.

This case highlights where classical machine learning performs best. The environment is stable, data is highly structured and generated continuously, and objectives remain consistent over time. Delivery patterns repeat, constraints are well defined, and optimisation targets are clear. In such conditions, predictive models reduce uncertainty and improve efficiency at scale in ways that are measurable and dependable.

The limitations of classical machine learning become visible when those conditions break down. During the onset of the COVID-19 pandemic, retail demand forecasting models across the world were trained on years of historical purchasing behaviour that assumed continuity in consumer patterns. When lockdowns began, demand shifted abruptly. Essentials experienced unpredictable surges, while other categories collapsed. Models continued to generate forecasts that appeared precise, but the underlying assumptions were no longer valid. Retailers were forced to override model outputs, rely on human judgement and adapt manually until sufficient new data allowed retraining.

This episode exposes a core boundary of classical machine learning: it is inherently backward-looking. It assumes that historical data contains reliable signals about the future. When structural breaks occur — whether from pandemics, regulatory shocks or disruptive innovation — the model cannot reason about events it has never encountered. Until new data is incorporated, its predictions may be confidently wrong.

There is an additional limitation revealed in these scenarios. Systems like ORION succeed because the data is structured and numerical. During periods of disruption, decision-makers must interpret qualitative uncertainty — policy announcements, news reports and behavioural shifts — much of which exists in unstructured form. Classical machine learning is not designed to interpret such context independently.

The leadership insight is therefore practical rather than theoretical. Classical machine learning excels in data-rich, stable environments with repeatable processes and clear optimisation goals. It delivers disciplined, scalable improvements that often remain invisible to customers but materially affect margins and efficiency. It does not anticipate novelty, interpret complex context or generate new

strategic responses. Organisations that deploy it deliberately, and remain cautious during periods of volatility, use it as a reliable decision-support engine rather than a source of misplaced confidence.

Video 4: Generative AI — From Prediction to Creation

For many years, the primary role of AI in enterprise settings was prediction. Organisations used machine learning systems to analyse historical data and estimate future outcomes — forecasting demand, assessing customer churn, or pricing financial risk. These systems improved decision-making, but their role was limited to answering structured questions about what was likely to happen next, based on patterns from the past.

Generative AI represents a clear shift in capability. Rather than estimating outcomes, it produces new outputs that can be used directly in everyday work. The focus moves from insight to creation — from understanding what might happen to actively generating content that helps move work forward. This distinction becomes concrete when looking at how generative systems are applied in real business contexts. A widely cited example is Coca-Cola, which in 2023 used generative AI in marketing campaigns that allowed both internal teams and consumers to create customised digital artwork aligned to brand assets. During parts of this initiative, more than 100,000 creative variations were reportedly generated, far beyond what traditional manual workflows could realistically support.

Historically, predictive models already played a role in marketing by identifying which audience segments were more likely to engage with certain content. That predictive layer remains valuable. What generative AI changed was the creative production process itself. Instead of relying on designers and copywriters to manually produce dozens of variations, generative tools enabled teams to generate hundreds or thousands of brand-aligned visuals and copy options rapidly, then test and refine them. Predictive AI might indicate that short, visually bold content performs well with a younger audience. Generative AI enables teams to immediately create multiple versions of such content across markets and formats, dramatically accelerating execution.

This distinction highlights the core difference between predictive and generative systems. Predictive AI produces probabilities, forecasts and scores that inform judgement. Generative AI produces artefacts — text, images, layouts, summaries or code — that can be used immediately. Its value lies in reducing the time, cost and effort required to create first drafts and explore alternatives. This pattern extends beyond marketing. In sales, predictive systems estimate the likelihood of deal closure, while generative tools draft tailored outreach emails and proposal outlines. In customer service, predictive models forecast ticket volumes, while generative systems draft responses, translate policies into plain language and suggest troubleshooting steps.

At the same time, the Coca-Cola example also reveals an important boundary. Generative AI can produce content at scale, but it does not understand brand reputation, cultural nuance or long-term strategic intent. Without guardrails, outputs can drift off-brand or become misaligned. Human oversight therefore remains essential. Creative leaders define constraints, review outputs and ensure consistency. Strategically, generative AI creates the most leverage in high-volume, iterative work where speed and experimentation matter. It does not replace predictive systems; it builds on them, turning insight into output and embedding AI directly into creative and knowledge workflows.

Video 5: Early Business Impact of Generative AI

For many emerging technologies, business impact remains abstract for years, discussed in pilots and conference talks without materially changing daily work. Generative AI is different. Its impact is already visible across organisations as an operational reality rather than a future promise. Teams are actively using it to create content, accelerate workflows and broaden participation in knowledge work. Its importance lies less in novelty and more in how rapidly it compresses time, lowers barriers to entry and reshapes how everyday tasks are performed.

One reason for this rapid adoption is accessibility. Unlike earlier AI systems that required complex integrations and specialist expertise, generative AI tools are available through simple, conversational interfaces. Employees can describe what they need in natural language and receive usable outputs almost instantly. This ease of use has driven rapid uptake across functions where documentation, communication and content creation are central. Marketing is one of the earliest areas of visible impact, with teams using tools such as Canva and Adobe Firefly to generate copy, layouts and visual concepts quickly, enabling faster creative exploration while maintaining brand safeguards.

Retail and e-commerce demonstrate a similar pattern. Platforms like Shopify have introduced generative features that help merchants create product descriptions, promotional emails and storefront content automatically. This allows small teams to scale their marketing output without relying on specialised copywriters. Beyond external-facing work, generative AI is also reshaping internal functions such as HR and learning. Teams use it to draft job descriptions, summarise policy updates, adapt training materials and outline learning content. While these outputs still require review, the reduction in initial drafting effort is significant.

Consulting and professional services offer another strong example of early impact. Much of consulting work involves research synthesis, structured reporting and executive communication. Firms including McKinsey have publicly discussed using generative AI to support these activities. In practice, the technology acts as a force multiplier. Consultants retain responsibility for judgement and client context, but spend less time on repetitive drafting and more time on analysis and insight generation, improving efficiency under tight timelines.

Across all these use cases, several strategic effects stand out. Generative AI compresses production time, enables faster experimentation, increases creative throughput and broadens participation beyond specialist roles. At the same time, its limitations are clear. These systems are not autonomous, do not understand organisational priorities, and can produce confident but incorrect or misaligned outputs. Human oversight, governance and domain expertise remain essential. The early business impact of generative AI is therefore best understood as augmentation rather than replacement — a practical tool that improves how work gets done today, while laying the foundation for more advanced capabilities tomorrow.

Video 6: What Changes When AI Creates Rather Than Predicts

Across the week, a clear shift has emerged in how artificial intelligence creates value in organisations. Traditional AI systems were primarily designed to predict outcomes by analysing historical data and

estimating what might happen next. These predictive systems remain deeply embedded in business operations, supporting decisions such as demand forecasting, churn prediction and risk assessment. Their role is to reduce uncertainty by producing probabilities, scores and forecasts that inform human judgement rather than replace it. Leaders remain responsible for interpreting these insights and deciding how to act.

Generative AI introduces a fundamentally different capability. Instead of estimating future outcomes, it produces new outputs in real time in response to intent and context. These outputs can include text, summaries, plans, designs or explanations that can be used directly in work. This shifts the role of AI from supporting decisions to supporting execution. Rather than asking “What is likely to happen?”, teams increasingly ask “What can be created now to move this task forward?”. As a result, generative AI becomes embedded in the flow of work itself, assisting with drafting, communication and coordination rather than operating only upstream in planning or analysis.

Comparing the two paradigms clarifies what has changed. Classical machine learning looks backward in order to look forward, relying heavily on structured historical data and stable patterns. Its strength lies in consistency and accuracy under known conditions. Generative AI, by contrast, works with prompts, context and intent. Its value comes from accelerating creation, expanding capacity and enabling rapid iteration. Importantly, this does not make the paradigms competitors. In most real-world workflows, they complement each other. Prediction helps determine where to focus attention, while creation helps translate that focus into action.

This complementarity is visible across functions. In marketing, predictive models help identify target segments and estimate likely responses, while generative tools create campaign copy and visual variations tailored to those segments. In sales, prediction prioritises leads and forecasts revenue, while generation supports engagement through drafted outreach and proposals. In operations and customer support, predictive systems help route and prioritise work, while generative systems assist with explanations, responses and documentation. In each case, prediction reduces uncertainty, while creation improves speed and alignment.

What ultimately changes when AI moves from prediction to creation is the point at which value is delivered. Predictive AI delivers value primarily at the decision stage. Generative AI delivers value at the execution stage, where outputs are visible and directly consumed. This brings new opportunities for productivity and experimentation, but also new responsibilities around review, governance and quality control. For leaders, the strategic question is not which paradigm to choose, but how to combine them effectively. Organisations that understand where prediction and creation each create leverage are better positioned to move beyond hype and apply AI in ways that are practical, responsible and genuinely valuable.

Video 7: Final Thoughts

This week has focused on clarifying a simple but frequently misunderstood idea: artificial intelligence is not a single capability. It is a set of distinct paradigms, each designed to address different types of problems and create value in different ways. Treating AI as one broad concept often leads to confusion, misplaced expectations, and poorly designed initiatives. Establishing these distinctions is therefore a necessary starting point for any meaningful business application of AI.

We began by reframing how AI should be understood in a business context. Rather than viewing it as a universal solution, we positioned AI as a progression of capabilities. Classical machine learning sits at the foundation of this progression. Its primary function is prediction. By analysing historical data, it helps organisations reduce uncertainty and improve decision-making. This includes forecasting demand, assessing risk, optimising pricing, and identifying patterns that would be difficult for humans to detect at scale. These systems are powerful, but they remain reactive, operating within the boundaries of past data and predefined objectives.

We then examined the strengths and limitations of classical machine learning more closely. While it improves accuracy, consistency, and efficiency, it does not create new material or act independently within workflows. It informs decisions, but it does not execute them. Recognising this boundary is essential. It explains why many AI initiatives delivered analytical insight without fundamentally changing how work was done. It also helps avoid unrealistic expectations about what predictive systems can and cannot achieve.

From there, we explored generative AI as the next shift. Generative systems move beyond prediction into creation. Instead of focusing on what is likely to happen, they support the production of content that can be used directly in work. This includes drafting text, generating ideas, summarising information, and supporting communication. As a result, generative AI fits naturally into everyday workflows, especially in areas where work begins with a blank page or requires repeated drafting and refinement. Its early business impact is already visible across functions such as marketing, HR, consulting, and internal operations, where speed and adaptability are critical.

At the same time, this week has emphasised that these paradigms do not replace one another. Prediction and creation are not competing approaches. In practice, they are often most effective when combined. Predictive systems help determine where attention should be focused and which actions are likely to matter most. Generative systems then help teams act on those insights by accelerating execution and communication. Together, they enable organisations to move more smoothly from insight to action.

The most important takeaway from this week is not about specific tools or technologies. It is about mindset. Leaders who understand these distinctions are better equipped to ask the right questions, make informed investment decisions, and design workflows that balance efficiency with human judgement.